TermMap: Mapping Clinical Mentions to ICD-10-CM and SNOMED CT

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Code, models and reproduction steps: `github.com/kingslayer35/HackLabs_IITR`

1 Problem and Approach

Each input is a clinical mention (for example "HTN" or "chronickidney dis stg 3"), usually with the note text around it. TermMap returns one ICD-10-CM code and one SNOMED CT US code for every input. Each system is scored by exact match, and the **headline score is the lower of the two accuracies**. Four facts about the data shaped the design:

9,646 labelled rows	**98k / 382k** ICD codes / SNOMED concepts	**1 in 4 / 1 in 2** ICD / SNOMED evaluation rows have a code never seen in training	**19% / 26%** accuracy of plain string matching (DEV)
Five entity types (diagnosis, procedure, lab, encounter, device) in one engine	A classifier over code IDs cannot work, so we search the whole dictionary	The long tail decides the score; memorising training labels is not enough	Notes use abbreviations, typos, category headers and wording that is not in the dictionary

2 Architecture and Innovations

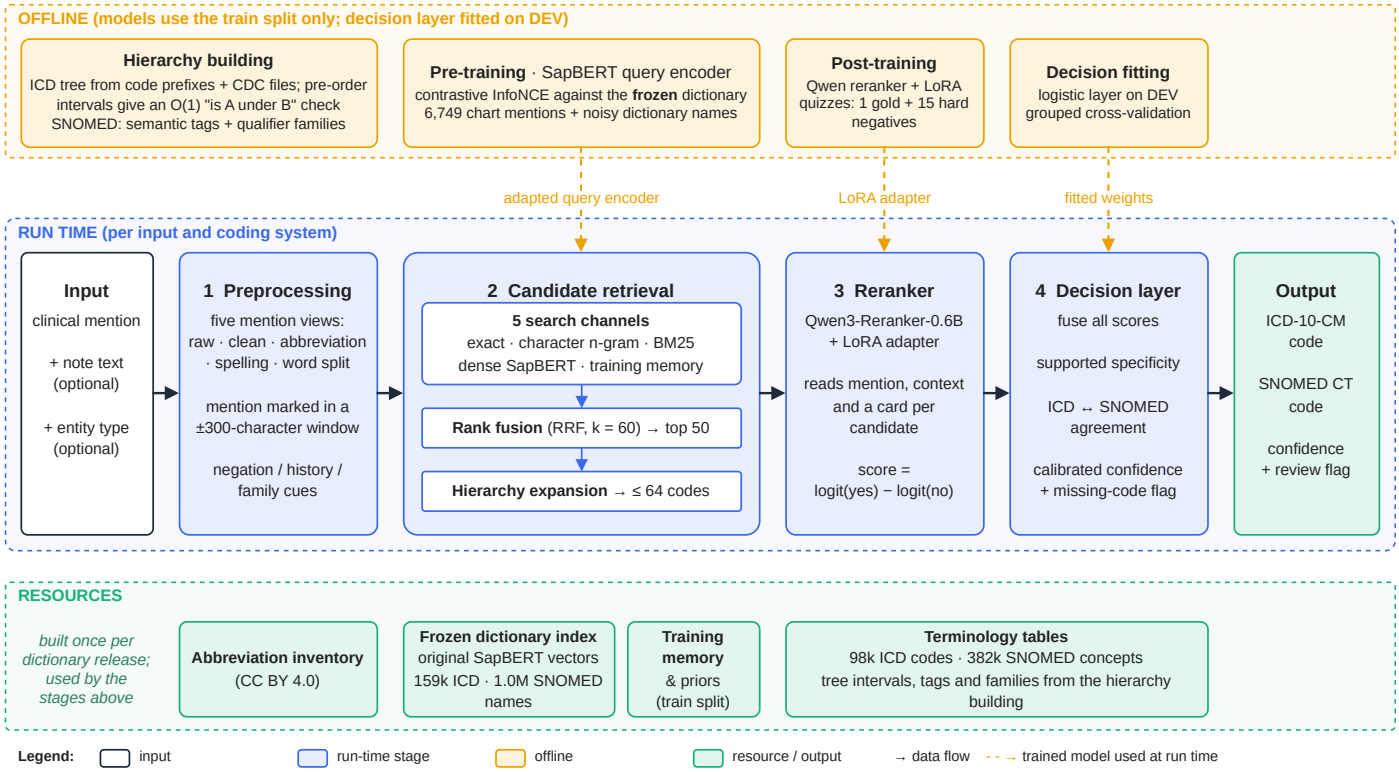


Figure 1: TermMap architecture. At run time each input goes through preprocessing, candidate retrieval over the full dictionary, the context-aware reranker and the decision layer, giving one legal code per system. Offline, we train two models (the SapBERT query encoder and a LoRA adapter on the Qwen reranker) and fit the decision layer on DEV. Shared resources are built once per dictionary release.

Key innovations.

- **Frozen-dictionary contrastive pre-training.** Only the query encoder learns, with InfoNCE over *every* concept of the system (98k ICD, 382k SNOMED) as negatives and the dictionary vectors kept fixed. Names shared between codes and the query's own name are masked out, and rare codes are up-weighted. Codes never seen in training keep their original vectors, so the adaptation cannot damage them. Our first, standard attempt moved the dictionary and lost recall.
- **Interval-indexed hierarchy without official relationship files.** The ICD tree is rebuilt from code prefixes and CDC files and stored in pre-order with an (in, out) interval per code. "Is a under b" is then $O(1)$ and any subtree is one contiguous slice. SNOMED concepts are grouped into qualifier "families" (left/right, acute/chronic, history of) by lexical rules. This drives candidate expansion and specificity.
- **A decision layer that decides, not just ranks.** Logistic fusion of all scores. It moves to a child code only when the text supports it, and it never emits an illegal ICD header (0 cases); training-supported legacy codes are the exception. It adds a soft ICD–SNOMED agreement bonus, a calibrated confidence and a separate "gold is missing" probability.
- **Reranker trained on its own retriever's mistakes.** Each LoRA quiz pairs the gold with 15 hard negatives taken from our retrieval pool (top misses and hierarchy siblings). The reranker sees structured candidate cards (description, parent, category, aliases), so it learns the confusions it will actually face.
- **Guarded multi-view preprocessing.** Five views of each mention. Clinical qualifiers (numbers, laterality, negation, stage) are protected, and abbreviations are expanded only with evidence in the note.

3 Method and Key Design Decisions

Table 1: Each stage, what it does and why. Details are in Appendices E–G.

Stage	What we do	Why
Preprocessing	Five mention views (raw, cleaned, abbreviation, spelling, word split); mention marked in a ±300-character window; negation/history/family cues flagged	A wrong "fix" changes the code: raw view always kept; numbers, </>, laterality, type, stage never edited; ambiguous abbreviations (MS, RA) left to the reranker
Hierarchy	ICD tree from code strings + CDC FY2026 files; SNOMED semantic tags and conservative lexical links	Supplied files have no parent links; the tree drives specificity and sibling candidates
Retrieval	Exact, character n-gram, BM25, dense and training-memory search; rank fusion (RRF, k=60); hierarchy siblings added (pool ≤64) [2]	Each channel catches different misses: typos (n-grams), synonyms (dense), local coding habits (memory)
Pre-training (SapBERT)	Contrastive training of the *query* encoder: pull a mention to its gold concept, push it from all other 98k ICD / 382k SNOMED concepts; dictionary vectors frozen [1]	Retrieval only finds what the embedding space places close; new wording and unseen codes need a well-shaped space. A frozen dictionary protects untrained codes
Post-training (Qwen LoRA)	Qwen3-Reranker-0.6B reads mention, context and a candidate card. LoRA (r=16, 1.7% of weights) trained on quizzes of 1 gold + 15 hard negatives [3, 4]	The base model knows no coding conventions and made ranking *worse* (TEST ICD top-1 40.8→32.2%); after post-training 58.3%
Decision layer	Logistic fusion, specificity rules, soft ICD–SNOMED agreement, calibrated confidence, missing-code probability [5]	No single score is reliable alone; fitted on DEV by grouped cross-validation, TEST untouched

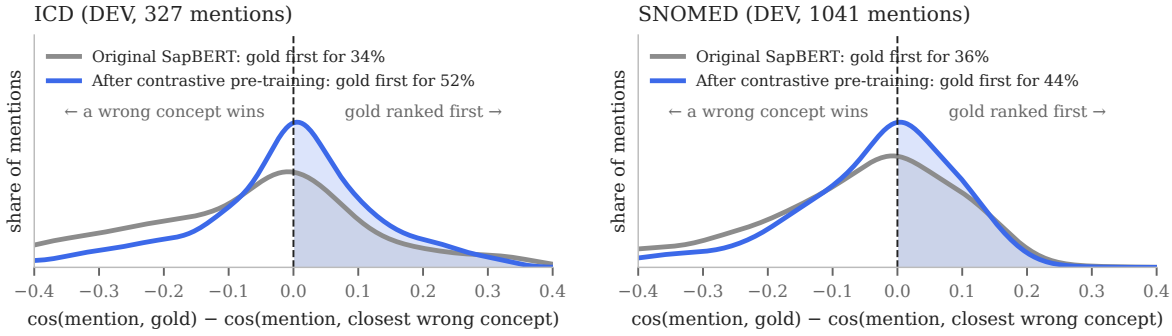


Figure 2: **Why pre-training matters** (DEV). For each mention: cos(gold) − cos(closest wrong concept); right of zero means the gold is ranked first. Contrastive pre-training shifts the mass right: gold-first rises 34→52% (ICD) and 36→44% (SNOMED), with the dictionary vectors unchanged. Per-mention movement is in Appendix E.

4 Results

Setup. The 9,646 labelled rows are split by note into **train 6,751 (70.0%)**, **DEV 1,368 (14.2%)** and **internal TEST 1,527 (15.8%)**, so one note never appears in two sets. Models, code frequencies and training memory use train only; decision layers are fitted on DEV and frozen before TEST is scored once. TEST is our own held-out split, not the

organisers' hidden set. Brackets: 95% grouped-bootstrap intervals.

Table 2: End-to-end results (exact-code accuracy, %). Baseline = full pipeline with original SapBERT and Qwen. Headline = lower of ICD and SNOMED (all types). SNOMED diagnoses = the DiagnosisCode subset. Recall@64 = gold among the retrieved candidates. **Bold** = best.

	Accuracy				Retrieval recall@64	
Pipeline	ICD	SNOMED	SNOMED diagnoses	**Headline**	ICD	SNOMED
DEV (ICD n=327, SNOMED n=1041)						
Baseline	52.9	43.1	51.8	43.1 [40.2, 46.2]	85.9	83.5
+ Pre-training	56.9	45.2	52.6	45.2 [42.0, 48.3]	**90.8**	**85.8**
+ Post-training	60.9	50.5	**57.8**	50.5 [47.6, 53.7]	85.9	83.5
+ Pre + Post (final)	**67.0**	**51.3**	57.3	**51.3** [48.1, 54.5]	**90.8**	**85.8**
Final − Baseline	+14.1	+8.2	+5.4	**+8.2** [5.7, 10.5]	+4.9	+2.3
TEST (ICD n=326, SNOMED n=1201)						
Baseline	53.1	42.7	51.5	42.7 [39.9, 45.7]	86.2	80.0
+ Pre-training	62.3	44.8	55.1	44.8 [42.0, 47.9]	**91.4**	**82.1**
+ Post-training	65.0	50.5	61.5	50.5 [47.6, 53.3]	86.2	80.0
+ Pre + Post (final)	**68.4**	**50.7**	**62.1**	**50.7** [48.1, 53.6]	**91.4**	**82.1**
Final − Baseline	+15.3	+8.0	+10.5	**+8.0** [5.6, 10.4]	+5.2	+2.1

Table 3: Long tail (20% of judging): final accuracy (%) by how often the gold code appears in training. Unseen = never, rare = 1–5 times, common = more than 5. **Bold** = best. Full tables with recall: Appendix B.

	ICD			SNOMED		
Pipeline	Unseen	Rare	Common	Unseen	Rare	Common
DEV (n = 81, 168, 78, 510, 468, 63)						
Baseline	33.3	62.5	52.6	28.2	57.9	54.0
+ Pre-training	39.5	65.5	56.4	29.4	60.0	63.5
+ Post-training	39.5	71.4	60.3	**36.7**	63.5	66.7
+ Pre + Post (final)	**46.9**	**73.8**	**73.1**	36.3	**65.2**	**69.8**
Final − Baseline	+13.6	+11.3	+20.5	+8.0	+7.3	+15.9
TEST (n = 77, 156, 93, 621, 494, 86)						
Baseline	42.9	57.1	54.8	29.3	55.7	65.1
+ Pre-training	51.9	66.7	63.4	31.6	58.1	64.0
+ Post-training	**57.1**	67.3	67.7	**39.5**	60.9	**69.8**
+ Pre + Post (final)	**57.1**	**73.7**	**68.8**	38.5	**63.2**	67.4
Final − Baseline	+14.3	+16.7	+14.0	+9.2	+7.5	+2.3

Table 4: Stage ablation of the final system. Green = gain over the previous stage. The pool (recall@64) is fixed by retrieval; later stages move the gold towards rank 1. All pipelines: Appendix A.

	Accuracy			ICD recall		SNOMED recall	
Stage	ICD	SNOMED	Headline	@10	@64	@10	@64
DEV							
Retrieval only	43.4	37.1	37.1	77.1	90.8	72.6	85.8
+ Reranker	54.4 (+11.0)	44.5 (+7.4)	44.5 (+7.4)	88.4	90.8	77.6	85.8
+ Decision layer	67.0 (+12.5)	51.3 (+6.8)	**51.3** (+6.8)	**89.3**	90.8	**81.2**	85.8
TEST							
Retrieval only	44.5	36.4	36.4	76.1	91.4	70.3	82.1
+ Reranker	53.4 (+8.9)	43.1 (+6.7)	43.1 (+6.7)	87.4	91.4	74.3	82.1
+ Decision layer	68.4 (+15.0)	50.7 (+7.6)	**50.7** (+7.6)	**89.9**	91.4	**78.3**	82.1

Verdict on the ablations (TEST).

- **Post-training is the largest single gain** (+7.7 headline). The original Qwen pushes the gold *down* (SNOMED top-1 36.7→29.1%); after LoRA it lifts it (44.0%).
- **Pre-training raises recall**: recall@64 ICD 86.2→91.4%, SNOMED 80.0→82.1%, and ICD accuracy +9.2. With post-training it still adds +3.4 ICD, but only +0.2 [−1.1, 1.9] headline, since SNOMED sets the headline.
- **The decision layer is the largest stage gain** (+15.0 ICD, +7.6 SNOMED): it repairs raw reranker mistakes and keeps every ICD code legal.

Table 5: Evidence for each judging criterion (TEST unless noted).

Criterion	Evidence
Accuracy (50%)	Headline 50.7% (DEV 51.3%), +8.0 [5.6, 10.4] over the baseline; ICD 68.4% (+15.3); SNOMED diagnoses 62.05%. Target 95% **not** met; SNOMED selection is the limit
Long tail (20%)	Table 3. Unseen codes: ICD 42.9→57.1%, SNOMED 29.3→38.5%; rare ICD codes 73.7%. Held-out dictionary names: ICD 15.1→67.0%, SNOMED 51.6→77.2% (App. B)
Engineering (15%)	One command (`bash run.sh in.csv out.csv`); fixed seeds; output validator; 1.57 s median per input for both systems incl. validation (p95 1.97 s) on one RTX A6000 (App. I)
Error analysis (15%)	Errors located by stage; calibration, missing-code stress test, worked cases (Sec. 5, App. D)

5 Error Analysis: Where It Still Fails

Table 6: Failure modes of the final system on TEST (103 ICD and 592 SNOMED errors). (a) splits errors by the pipeline stage where they arise. (b) splits the same errors by how far the prediction is from the gold.

Failure mode	ICD	SNOMED	Main cause
(a) Where in the pipeline			
Gold never retrieved (not in the 64 candidates)	28	215	Chart wording far from every dictionary name; for SNOMED, no official hierarchy to expand a near hit to its parent or child
Right concept, wrong specificity	11	6	Text does not clearly support the more (or less) specific code
Gold retrieved, another code picked	64	371	Several near-identical concepts; local coding conventions
(b) How far the prediction is from the gold			
Same ICD category / SNOMED near miss	29	10	Sibling code: right condition, wrong detail
Same SNOMED semantic tag	–	254	Near-duplicate concepts, e.g. gold "Estimated creatinine clearance" vs predicted "Measurement of renal clearance of creatinine" (both procedures)
Same SNOMED top level, other tag	–	43	Disorder vs finding, e.g. "Uncontrolled type 2 diabetes mellitus" vs "Diabetic - poor control"
Different ICD category / SNOMED top level	74	285	Wrong concept type, e.g. procedure vs observable entity, or wrong target in the sentence

Why it still fails. Extended error analysis (all pipelines, error classes, worked cases, calibration and a missing-code stress test) is in Appendix D.

- **Missing SNOMED hierarchy.** The supplied SNOMED file has no IS-A relationships, and the official graph needs a UMLS licence. Without it we cannot expand a near hit to its real parent or children, cannot use the graph to tell a general concept from a specific one, and cannot measure IS-A distance. 254 of the SNOMED errors share the gold's semantic tag; many of these are concepts the graph would separate.
- **Choosing, not finding, is the main loss.** The gold is in the final top 10 for 89.9% of ICD and 78.3% of SNOMED inputs (recall@10), and in the 64 candidates for 91.4% / 82.1%, yet it is ranked first for only 68.4% / 50.7%. The right code is usually retrieved; it is just not chosen. When the gold is retrieved, the right code is picked 74.8% of the time for ICD and 61.8% for SNOMED. Procedures and lab services are hardest.
- **Concept-type confusion.** 285 SNOMED errors fall in a different top-level hierarchy, for example a procedure where the label is an observable entity. The entity type helps, but it is only a soft prior.
- **Unseen codes.** Accuracy on never-seen SNOMED codes is 38.5%, against 63–67% on rare and common ones. Training examples cannot teach these codes.

Challenges and learnings.

- **Pre-training v1 was abandoned:** standard SapBERT-style metric learning also moved the dictionary vectors (cosine to the original 0.43 for ICD) and full-pipeline recall@64 fell 84.1→81.7%. v2 freezes the dictionary and trains only queries.
- **Aggressive cleaning hurt**, so abbreviation expansions are evidence-gated and the raw view is always kept. **Lower loss ≠ higher accuracy**: LoRA epoch 2 was chosen over the automatic epoch-1 pick.

6 Future Work and Expected ROI

Alternative solution already explored. Alongside the main engine we built a tool-using LLM agent. It searches the dictionaries, reads code descriptions and ICD relatives, looks at similar training mentions and then commits to a validated code. On 500 diagnosis inputs it reached 70.6% ICD and 71.4% SNOMED, about 60% on unseen codes (Appendix H). It needs several hosted LLM calls per input and is not deterministic, so on its own it is a second opinion rather than a replacement. Our next step is to combine the two: for the hardest, low-confidence cases, an agentic loop will call our retrieval engine as a tool after reranking. It can search again, inspect candidates and check the reranked choice, which

should raise confidence in the final code.

Table 7: Next steps, ordered by expected return for the effort. Latency details: Appendix I.

Action	Why / expected impact	Impact	Effort
Better SNOMED selection for procedures and labs	377 of 592 SNOMED errors already have the gold among the candidates; targeted quizzes or a larger reranker (Qwen3-Reranker-4B) attack the largest bucket	High	Medium
UMLS licence: SNOMED IS-A graph and official SNOMED→ICD map	Real parent/child expansion, IS-A distance metrics, a stronger ICD–SNOMED agreement signal	High	Low
Faster lexical scoring	Character n-gram and BM25 scoring take 56.5% of request time; no accuracy change expected	High (speed)	Low
Agentic loop on top of the retrieval engine	For low-confidence inputs only, the agent (Appendix H, about 71% on diagnoses) calls our retrieval engine as a tool after reranking to re-check the final code; cost stays bounded	Medium	Medium
Stronger missing-code detection	Makes the review flag useful for human review (catches only 8–13% today)	Medium	Medium
Retrain on all labelled data; test masked-LM pre-training	43% more supervision for the deployed models; check whether domain pre-training adds to contrastive adaptation	Medium	Low

7 Team Contributions

Member	Contribution
Garv Shailesh Agarwal	Preprocessing; terminology layer and hierarchy; candidate retrieval; edge cases
Pratham Singla	SapBERT contrastive pre-training; Qwen reranker post-training
Adesh Gupta	Decision layer; agentic retrieval pipeline

References

[1] F. Liu, E. Shareghi, Z. Meng, M. Basaldella and N. Collier. Self-Alignment Pretraining for Biomedical Entity Representations. *NAACL*, 2021. https://aclanthology.org/2021.naacl-main.334/

[2] Tao and Liu. Neighborhood-Aware Dual Biomedical Entity Linking (PILOT). Preprint, 2026. https://arxiv.org/abs/2608.04144

[3] Wang et al. BioELX: Cross-lingual Biomedical Entity Linking via Alias-based Retrieval and LLM Ranking. Preprint, 2026. https://arxiv.org/abs/2605.27380

[4] Kadusabe et al. Context-Aware SNOMED CT Entity Linking for Clinical Text. *Clinical NLP Workshop*, 2026. https://lrec.elra.info/lrec2026-ws-clinicalnlp-33

[5] Jiang et al. One Score, Two Decisions: Selective Prediction on the Rare-Disease Tail. Preprint, 2026. https://arxiv.org/abs/2608.14683

Appendices

A Detailed Accuracy Results and Hierarchical Diagnostics

Every pipeline has three stages: *retrieval only* (fused candidate ranking), *+ reranker* (raw Qwen order) and *+ decision layer* (final output). Accuracy equals recall@1. Recall@k is the share of inputs whose gold code is in the first k ranked codes.

Table 8: All pipelines and stages on **DEV** (ICD n=327, SNOMED n=1,041). The DEV decision layers use out-of-fold (nested grouped cross-validation) estimates.

	ICD				SNOMED (all types)				
Stage	Accuracy	R@10	R@32	R@64	Accuracy	R@10	R@32	R@64	**Headline**
Baseline									
Retrieval only	39.4 [34.0, 44.9]	**67.3**	79.2	85.9	36.8 [34.0, 39.5]	**69.2**	79.2	83.5	36.8
+ Reranker	32.4 [27.3, 37.5]	**69.7**	81.7	85.9	30.1 [27.0, 32.9]	**69.2**	81.2	83.5	30.1
+ Decision layer	**52.9 [47.5, 58.4]**	**81.3**	**85.6**	**85.9**	**43.1 [40.0, 46.1]**	**76.9**	**82.6**	**83.5**	**43.1**
+ Pre-training									
Retrieval only	43.4 [37.5, 49.1]	**77.1**	87.8	90.8	37.1 [34.1, 39.9]	**72.6**	82.8	85.8	37.1
+ Reranker	31.2 [26.3, 36.4]	**71.9**	85.9	90.8	30.3 [27.2, 33.2]	**69.1**	82.7	85.8	30.3
+ Decision layer	**56.9 [51.5, 62.2]**	**86.9**	**90.8**	**90.8**	**45.2 [42.1, 48.4]**	**80.0**	**84.9**	**85.8**	**45.2**
+ Post-training									
Retrieval only	39.4 [34.0, 44.9]	**67.3**	79.2	85.9	36.8 [34.0, 39.5]	**69.2**	79.2	83.5	36.8
+ Reranker	55.7 [50.3, 61.2]	**84.7**	85.9	85.9	45.4 [42.3, 48.5]	**76.9**	83.0	83.5	45.4
+ Decision layer	**60.9 [55.7, 65.8]**	**84.4**	**86.2**	**86.2**	**50.5 [47.6, 53.5]**	**79.2**	**83.3**	**83.5**	**50.5**
+ Pre + Post (final)									
Retrieval only	43.4 [37.5, 49.1]	**77.1**	87.8	90.8	37.1 [34.1, 39.9]	**72.6**	82.8	85.8	37.1
+ Reranker	54.4 [49.2, 60.1]	**88.4**	90.8	90.8	44.5 [41.1, 47.7]	**77.6**	85.2	85.8	44.5
+ Decision layer	**67.0 [61.8, 71.8]**	**89.3**	**90.8**	**90.8**	**51.3 [48.2, 54.5]**	**81.2**	**85.5**	**85.8**	**51.3**

Table 9: All pipelines and stages on **internal TEST** (ICD n=326, SNOMED n=1,201). Decision layers are frozen from DEV.

	ICD				SNOMED (all types)				
Stage	Accuracy	R@10	R@32	R@64	Accuracy	R@10	R@32	R@64	**Headline**
Baseline									
Retrieval only	40.8 [35.6, 45.5]	**69.9**	81.3	86.2	36.7 [33.7, 39.5]	**67.1**	75.4	80.0	36.7
+ Reranker	32.2 [26.9, 37.2]	**71.2**	85.3	86.2	29.1 [26.5, 31.6]	**65.4**	77.5	80.0	29.1
+ Decision layer	**53.1 [47.4, 58.0]**	**84.0**	**86.5**	**86.5**	**42.7 [39.8, 45.4]**	**74.5**	**79.4**	**80.0**	**42.7**
+ Pre-training									
Retrieval only	44.5 [39.3, 50.0]	**76.1**	87.7	91.4	36.4 [33.4, 39.1]	**70.3**	78.5	82.1	36.4
+ Reranker	31.3 [26.0, 36.4]	**74.8**	88.3	91.4	29.2 [26.8, 31.6]	**66.6**	78.9	82.1	29.2
+ Decision layer	**62.3 [56.9, 67.1]**	**86.5**	**91.4**	**91.4**	**44.8 [41.7, 47.5]**	**77.4**	**81.4**	**82.1**	**44.8**
+ Post-training									
Retrieval only	40.8 [35.6, 45.5]	**69.9**	81.3	86.2	36.7 [33.7, 39.5]	**67.1**	75.4	80.0	36.7
+ Reranker	58.3 [53.1, 63.1]	**83.1**	86.2	86.2	44.0 [41.2, 46.8]	**73.9**	79.4	80.0	44.0
+ Decision layer	**65.0 [60.0, 69.8]**	**85.6**	**86.5**	**86.5**	**50.5 [47.4, 53.1]**	**76.9**	**79.8**	**80.0**	**50.5**
+ Pre + Post (final)									
Retrieval only	44.5 [39.3, 50.0]	**76.1**	87.7	91.4	36.4 [33.4, 39.1]	**70.3**	78.5	82.1	36.4
+ Reranker	53.4 [47.8, 58.4]	**87.4**	90.8	91.4	43.1 [40.4, 45.7]	**74.3**	81.3	82.1	43.1
+ Decision layer	**68.4 [63.4, 73.4]**	**89.9**	**91.4**	**91.4**	**50.7 [47.9, 53.4]**	**78.3**	**81.8**	**82.1**	**50.7**

Reading. The final system gets the right ICD category for 77–78% of inputs and the right SNOMED semantic tag

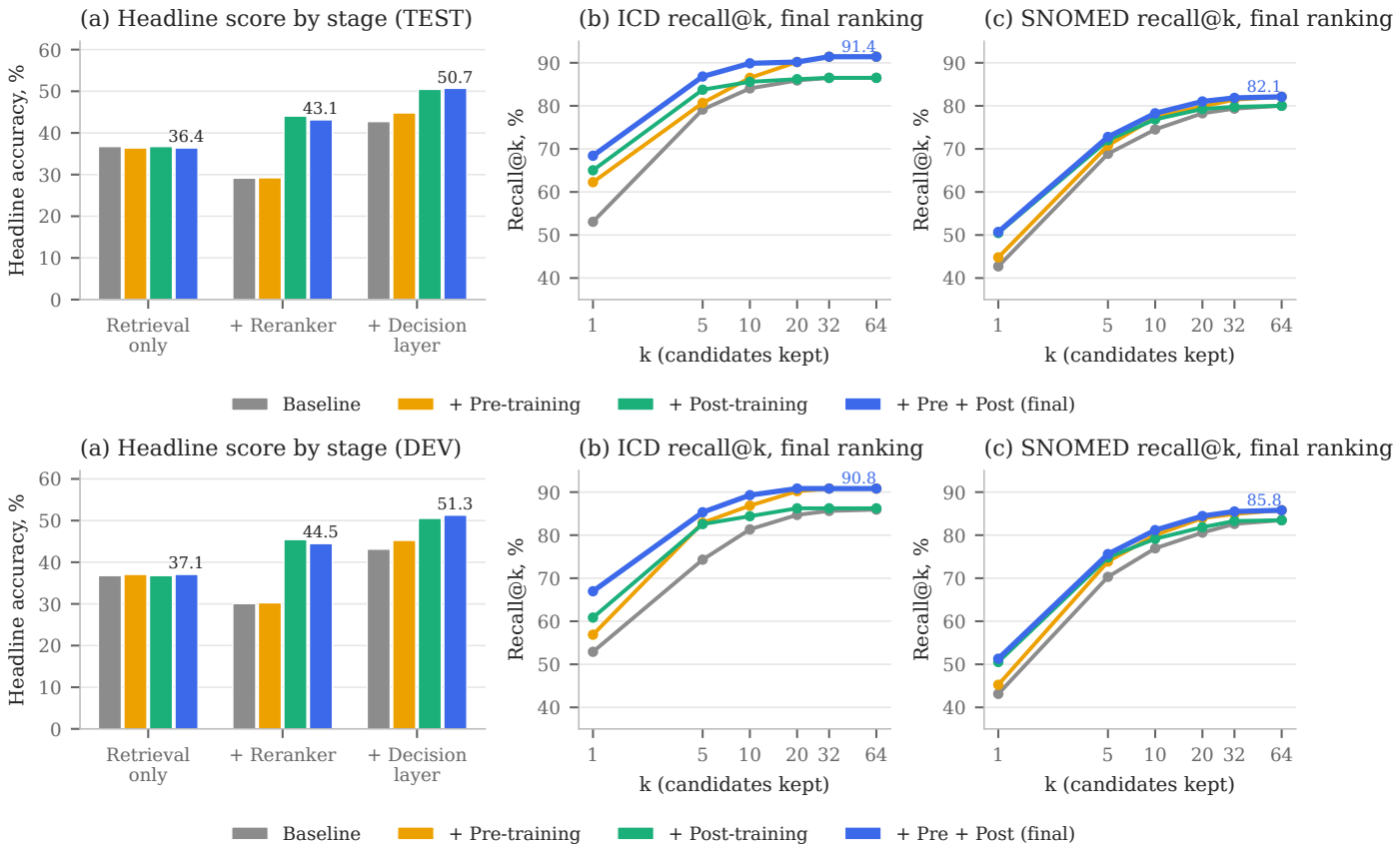


Figure 3: Top: TEST. Bottom: DEV. (a) Headline accuracy after each stage. (b, c) Recall@k of the final ranking; k=1 is the accuracy.

Table 10: Paired differences in final accuracy (percentage points, 95% paired grouped-bootstrap interval).

Comparison	ICD	SNOMED	Headline
DEV			
+ Post-training vs Baseline	+7.95 [3.42, 12.61]	+7.40 [5.12, 9.71]	+7.40 [5.12, 9.71]
+ Pre-training vs Baseline	+3.98 [−0.31, 7.99]	+2.11 [0.38, 3.82]	+2.11 [0.38, 3.82]
+ Pre + Post (final) vs Baseline	**+14.07 [8.96, 19.18]**	**+8.17 [5.74, 10.52]**	**+8.17 [5.74, 10.52]**
+ Pre + Post (final) vs + Post-training	+6.12 [2.42, 10.29]	+0.77 [−1.02, 2.40]	+0.77 [−1.02, 2.40]
+ Pre + Post (final) vs + Pre-training	+10.09 [5.55, 14.54]	+6.05 [4.05, 8.04]	+6.05 [4.05, 8.04]
TEST			
+ Post-training vs Baseline	+11.96 [7.37, 16.72]	+7.74 [5.36, 9.97]	+7.74 [5.36, 9.97]
+ Pre-training vs Baseline	+9.20 [5.76, 12.93]	+2.08 [0.58, 3.63]	+2.08 [0.58, 3.63]
+ Pre + Post (final) vs Baseline	**+15.34 [11.04, 20.69]**	**+7.99 [5.61, 10.37]**	**+7.99 [5.61, 10.37]**
+ Pre + Post (final) vs + Post-training	+3.37 [0.00, 6.96]	+0.25 [−1.13, 1.92]	+0.25 [−1.13, 1.92]
+ Pre + Post (final) vs + Pre-training	+6.13 [2.50, 10.29]	+5.91 [3.74, 7.92]	+5.91 [3.74, 7.92]

Table 11: Inputs labelled in both systems: share with *both* codes correct (%).

Pipeline	n	Both correct
DEV		
Baseline	307	34.85 [29.67, 40.19]
+ Pre-training	307	39.74 [34.51, 45.14]
+ Post-training	307	40.72 [35.20, 46.41]
+ Pre + Post (final)	307	46.58 [41.17, 52.34]
TEST		
Baseline	300	37.67 [32.11, 42.60]
+ Pre-training	300	43.67 [37.84, 49.64]
+ Post-training	300	50.33 [43.93, 56.21]
+ Pre + Post (final)	300	51.67 [45.67, 57.34]

Table 12: Hierarchical (partial-credit) diagnostics of the final outputs (%). ICD category = correct 3-character category. SNOMED tag / top level = correct semantic tag / top-level hierarchy. Same core = same concept apart from qualifiers. Over/under = predicted code more/less specific than gold. Illegal headers = non-billable ICD headers emitted outside training-supported exceptions.

Pipeline	ICD category	SN tag	SN top level	SN same core	ICD over / under	SN over / under	Illegal headers
DEV							
Baseline	66.97	66.76	71.57	44.38	1.22 / 2.14	0.10 / 0.38	0
+ Pre-training	71.56	68.59	72.91	46.88	2.45 / 1.83	0.19 / 0.58	0
+ Post-training	74.01	71.76	75.60	51.78	1.22 / 1.22	0.10 / 0.29	0
+ Pre + Post (final)	77.98	73.49	77.14	52.74	1.22 / 1.53	0.10 / 0.29	0
TEST							
Baseline	65.64	65.03	69.94	43.88	1.84 / 3.99	0.42 / 0.33	0
+ Pre-training	71.78	67.86	72.44	46.13	1.84 / 2.45	0.33 / 0.33	0
+ Post-training	73.31	71.19	75.35	51.46	1.53 / 0.92	0.25 / 0.17	0
+ Pre + Post (final)	77.30	72.52	76.10	51.62	1.84 / 1.53	0.25 / 0.25	0

for about 73%, well above its exact accuracy. Many errors land in the right area with the wrong concept. Over- and under-coding stay below 2% (ICD) and 0.3% (SNOMED). No illegal header code is ever emitted. True SNOMED IS-A distance cannot be reported without the licensed RF2 release.

B Long-Tail Evaluation

Buckets use **training-split** frequency of the gold code: *unseen* = 0 occurrences, *rare* = 1–5, *common* = more than 5. About half of all SNOMED evaluation rows are unseen, so this bucket drives the headline.

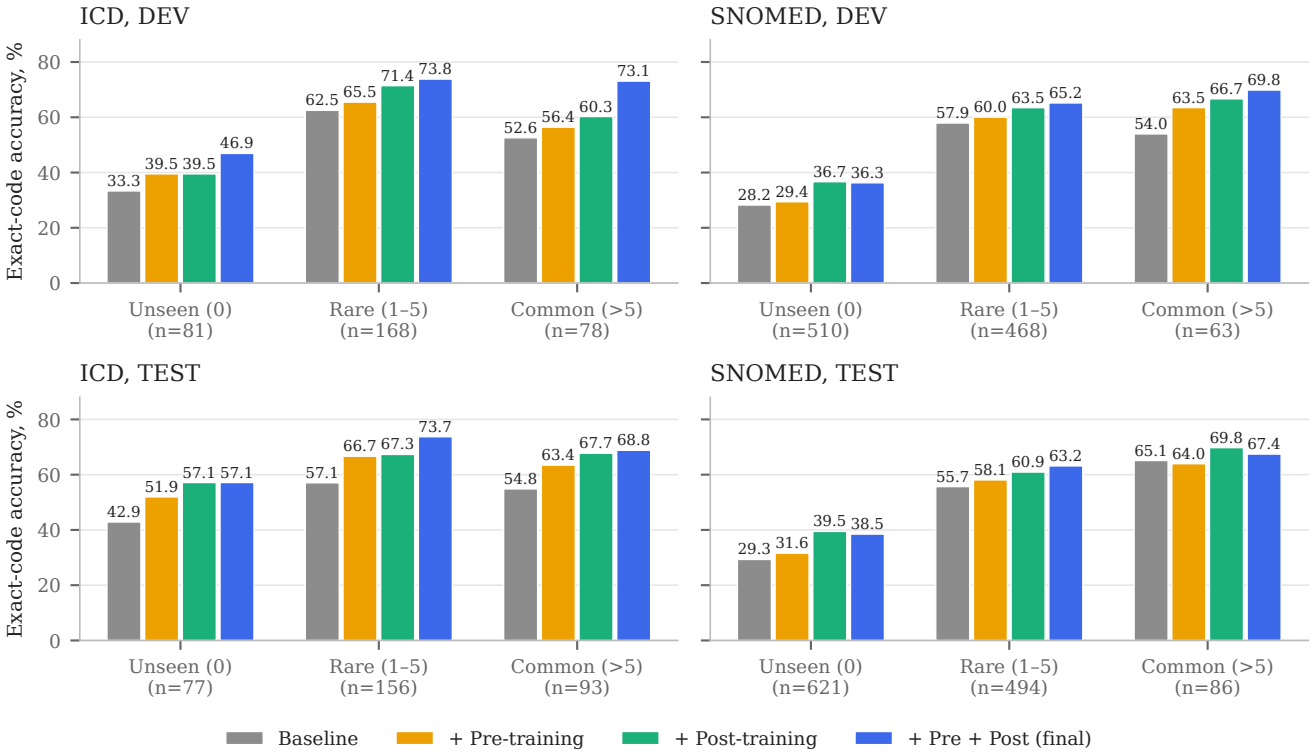


Figure 4: Final accuracy by code frequency for all four pipelines. Gains are largest on unseen and rare codes.

Held-out dictionary probe (codes and names never seen in training)

For 1,500 codes per system we withheld one dictionary name from every index and from training. We then query with that name, either clean or corrupted (typos, merged words, truncation, word inversion). This tests whether the engine can map wording that it has never seen, as in the long tail. The probe is separate from the chart sets.

Reading. Post-training brings the largest probe gain (ICD +45.9). The original reranker cannot match an unseen dictionary name to its code. Pre-training adds +6.1 ICD on top. The probe includes 136 ICD target headers that the specificity policy will not emit, so the ICD ceiling under our output rule is 90.9%.

Table 13: Baseline vs final system by bucket. Incoming R@64 = gold in the retrieval pool. Selection = accuracy among inputs whose gold was in that pool. Category/tag = correct ICD category or SNOMED semantic tag.

Pipeline	System	Bucket	n	Final accuracy	Incoming R@64	Selection given gold in pool	ICD category / SN tag
DEV							
Baseline	ICD	unseen	81	33.33 [21.69, 44.58]	76.54 [66.26, 85.19]	43.55 [30.51, 56.26]	54.32 [43.21, 65.39]
Baseline	ICD	rare	168	**62.50** [55.62, 69.50]	91.67 [87.57, 95.71]	**68.18** [61.04, 75.16]	73.21 [66.66, 79.88]
Baseline	ICD	common	78	52.56 [41.77, 62.96]	83.33 [73.97, 91.03]	**63.08** [50.77, 75.00]	66.67 [55.26, 76.47]
Baseline	SNOMED	unseen	510	28.24 [24.45, 32.39]	77.65 [73.89, 81.31]	36.36 [31.37, 41.22]	56.27 [51.43, 60.71]
Baseline	SNOMED	rare	468	57.91 [53.29, 62.67]	89.32 [86.37, 91.96]	**64.83** [60.10, 69.54]	76.50 [72.69, 80.72]
Baseline	SNOMED	common	63	53.97 [42.86, 65.57]	87.30 [78.46, 95.24]	**61.82** [50.00, 74.55]	79.37 [68.85, 88.53]
+ Pre + Post (final)	ICD	unseen	81	46.91 [35.00, 58.98]	81.48 [71.95, 90.00]	57.58 [45.31, 69.84]	59.26 [48.72, 70.51]
+ Pre + Post (final)	ICD	rare	168	**73.81** [67.27, 80.36]	93.45 [89.65, 96.91]	**78.98** [72.33, 85.53]	83.93 [78.12, 89.41]
+ Pre + Post (final)	ICD	common	78	**73.08** [62.81, 82.50]	94.87 [89.33, 98.75]	**77.03** [67.53, 85.53]	84.62 [76.62, 92.41]
+ Pre + Post (final)	SNOMED	unseen	510	36.27 [31.85, 40.64]	78.24 [74.40, 81.80]	46.37 [41.22, 51.20]	63.73 [58.87, 68.08]
+ Pre + Post (final)	SNOMED	rare	468	**65.17** [60.77, 69.80]	92.95 [90.44, 95.29]	**70.11** [65.31, 74.71]	81.41 [77.80, 84.97]
+ Pre + Post (final)	SNOMED	common	63	**69.84** [59.02, 80.00]	93.65 [87.10, 98.48]	**74.58** [64.91, 85.00]	93.65 [87.30, 98.48]
TEST							
Baseline	ICD	unseen	77	42.86 [32.43, 54.29]	76.62 [66.26, 86.11]	55.93 [42.86, 69.23]	68.83 [58.44, 78.38]
Baseline	ICD	rare	156	57.05 [48.38, 65.00]	88.46 [82.91, 93.79]	**63.77** [55.48, 71.74]	67.31 [59.03, 74.50]
Baseline	ICD	common	93	54.84 [44.44, 65.27]	90.32 [84.21, 95.70]	**60.71** [50.62, 70.93]	60.22 [50.00, 70.12]
Baseline	SNOMED	unseen	621	29.31 [25.32, 32.83]	72.79 [68.90, 75.97]	40.27 [35.34, 44.70]	57.00 [53.22, 60.96]
Baseline	SNOMED	rare	494	55.67 [51.38, 60.32]	87.65 [84.75, 90.47]	**63.51** [59.23, 67.77]	73.08 [68.71, 77.68]
Baseline	SNOMED	common	86	**65.12** [55.43, 76.47]	88.37 [81.61, 95.06]	**73.68** [63.75, 83.33]	76.74 [67.67, 86.25]
+ Pre + Post (final)	ICD	unseen	77	57.14 [45.83, 67.57]	81.82 [72.22, 90.54]	**69.84** [56.92, 81.16]	71.43 [60.87, 81.93]
+ Pre + Post (final)	ICD	rare	156	**73.72** [66.19, 80.63]	95.51 [91.67, 98.73]	**77.18** [70.55, 83.34]	83.33 [76.25, 89.61]
+ Pre + Post (final)	ICD	common	93	**68.82** [58.69, 77.89]	92.47 [86.81, 96.94]	**74.42** [64.70, 84.53]	72.04 [62.62, 81.11]
+ Pre + Post (final)	SNOMED	unseen	621	38.49 [33.76, 42.21]	73.11 [69.23, 76.40]	52.64 [47.34, 56.77]	63.93 [60.04, 67.52]
+ Pre + Post (final)	SNOMED	rare	494	**63.16** [58.79, 67.51]	91.50 [88.76, 93.65]	**69.03** [64.71, 73.08]	81.17 [77.59, 84.42]
+ Pre + Post (final)	SNOMED	common	86	**67.44** [52.88, 81.58]	93.02 [87.88, 97.59]	**72.50** [59.34, 86.67]	84.88 [77.19, 92.41]

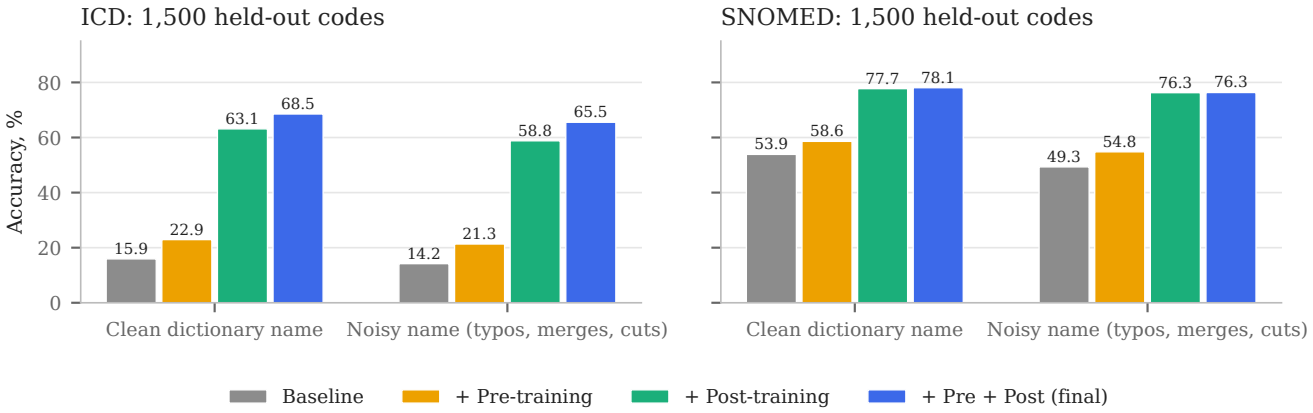


Figure 5: Probe accuracy for clean and noisy held-out names.

Table 14: Probe results (3,000 queries per system) and paired differences.

Pipeline	System	Accuracy	Clean	Noisy	R@10	R@64
Baseline	ICD	15.07 [13.43, 16.87]	15.93	14.20	52.03	84.53
Baseline	SNOMED	51.60 [49.00, 53.90]	53.87	49.33	88.03	91.87
+ Pre-training	ICD	22.13 [20.20, 24.23]	22.93	21.33	71.67	85.80
+ Pre-training	SNOMED	56.70 [54.30, 58.90]	58.60	54.80	88.63	91.90
+ Post-training	ICD	**60.97** [58.90, 63.24]	**63.13**	58.80	82.87	85.20
+ Post-training	SNOMED	**77.00** [75.07, 78.97]	**77.73**	**76.27**	90.37	91.87
+ Pre + Post (final)	ICD	**67.03** [65.03, 69.27]	**68.53**	**65.53**	84.30	86.03
+ Pre + Post (final)	SNOMED	**77.20** [75.23, 79.07]	**78.07**	**76.33**	90.63	91.90

Comparison	ICD	SNOMED	Headline
+ Post-training vs Baseline	+45.90 [43.57, 48.36]	+25.40 [23.41, 27.48]	+45.90 [43.57, 48.36]
+ Pre-training vs Baseline	+7.07 [5.65, 8.53]	+5.10 [3.76, 6.58]	+7.07 [5.65, 8.53]
+ Pre + Post (final) vs Baseline	**+51.97 [49.76, 54.27]**	**+25.60 [23.59, 27.69]**	**+51.97 [49.76, 54.27]**
+ Pre + Post (final) vs + Post-training	+6.07 [4.80, 7.43]	+0.20 [−0.62, 0.95]	+6.07 [4.80, 7.43]
+ Pre + Post (final) vs + Pre-training	+44.90 [42.55, 47.24]	+20.50 [18.51, 22.45]	+44.90 [42.55, 47.24]

C Engineering and Reproducibility

Running it. After the private artifacts are restored and the indexes are rebuilt, one command produces the submission file:

```
bash run.sh /abs/input.csv /abs/predictions.csv
```

It runs a smoke test, predicts both systems and validates the output. Input: `chunk_id, clinical_entity` (`chunk_text` and `subconcept` optional). Output, in the same order: `chunk_id, clinical_entity, icd10cm_code, snomed_code`. Codes stay strings, so long SNOMED IDs never become floats. A diagnostics file stores the top candidates, the confidence and the review flag.

Determinism. Every random step uses seed 13. We set `torch.use_deterministic_algorithms`, `CUBLAS_WORKSPACE_CONFIG=:4096:8`, fixed batch order, scores rounded to 10^{-5} and ties broken by code string. vLLM runs with batch-invariant kernels. No API key is needed, and no data leaves the machine.

Table 15: Hardware (one GPU server).

Component	Specification and use
GPUs	3 × NVIDIA RTX A6000 (48 GB, Ampere, sm_86); 1 × NVIDIA RTX PRO 5000 Blackwell (48 GB, sm_120)
CPU / RAM	36 cores, Intel Xeon Platinum 8352V; 225 GB RAM
Pre-training	1 × RTX A6000; 1,590 steps in about 11 minutes, plus evaluation every 50 steps
Post-training	1 × RTX A6000; about 2.5 hours per LoRA epoch (2 epochs)
Evaluation	DEV baseline/+pre: vLLM on Blackwell. DEV +post/final: vLLM on A6000. TEST baseline/+pre: Transformers on A6000. TEST +post/final: vLLM on Blackwell. Backends use bf16; small numerical differences are possible
Latency benchmark	Final system, vLLM, 1 × RTX A6000
Software	Python 3.11; torch 2.8.0+cu128; transformers 4.56.2; peft 0.17.1; sentence-transformers 5.1.1; bm25s 0.2.14; scikit-learn 1.7.2; serving: vLLM 0.19.0 (separate lockfile)

Table 16: Pre-training (SapBERT, P2-v2).

Base model	`cambridgeltl/SapBERT-from-PubMedBERT-fulltext`
Trained part	Query encoder only, one per system; dictionary vectors frozen
Loss	InfoNCE over all concepts, $\tau = 0.03$
Optimizer	AdamW, lr 2e-5, weight decay 0.01 (not on bias/LayerNorm)
Schedule	10% warmup, linear decay
Batch / epochs	64 / max 10 (159 steps per epoch)
Max length / pooling	40 tokens / normalised [CLS]
Precision / clip	bf16 / gradient norm 1.0
Data per epoch	6,749 chart mentions + 3,374 dictionary-name queries (about 40% noised)
Weights	rare codes $1/\sqrt{f}$, mean 1
Selection	lowest DEV loss per system, checked every 50 steps, patience 6
Selected	ICD step 1,431 (epoch 9.0); SNOMED step 950 (epoch 6.0)

Table 17: Post-training (Qwen reranker LoRA).

Base model	`Qwen/Qwen3-Reranker-0.6B`
LoRA	r 16, α 32, dropout 0.05; q, k, v, o, gate, up, down projections
Trainable	10.09M of 605.87M (1.7%)
Loss	softmax cross-entropy over 16 candidates (yes − no logit)
Optimizer	AdamW, lr 1e-4
Schedule	5% warmup, cosine decay
Batch / epochs	8 quizzes per update / 2
Max length	512 tokens
Precision / clip	bf16 / gradient norm 1.0
Data per epoch	10,502 quizzes = 168,032 candidate judgments
Selection	epoch 2 (explicit choice; epoch 1 kept)
Serving	vLLM 0.19.0, bf16, max length 512, Triton attention, eager mode

Latency and footprint. Median 1.57 s per input for both systems (p95 1.97 s), measured on one RTX A6000 including output validation; peak GPU memory 39.5 GiB. The component breakdown is in Appendix I.

Not used: UMLS, SNOMED RF2 relationship files, the official SNOMED→ICD map, and any external LLM API.

Table 18: External resources (all named, all used locally).

Resource	Use	Licence
Supplied ICD-10-CM and SNOMED CT US workbooks	Legal output codes and descriptions	Challenge-provided
CDC ICD-10-CM FY2026 Tabular and Index XML	Chapters, blocks, specificity notes	U.S. public domain
Clinical Abbreviation Meta-Inventory	Abbreviation candidates	CC BY 4.0
SapBERT-from-PubMedBERT-fulltext	Dictionary encoder; base of adapted query encoders	Apache-2.0
Qwen3-Reranker-0.6B	Reranker base for LoRA	Apache-2.0
SNOMED CT semantic tag list (Editorial Guide, Jan 2026)	Tag → top-level mapping	Public documentation
simple-icd-10-cm	ICD hierarchy cross-check	MIT

D Extended Error Analysis and Coverage

Table 19: Error location for every pipeline. Rescues/demotions count inputs whose top-1 became right/wrong at that stage.

Pipeline	System	Gold not retrieved	Wrong specificity	Wrong pick	Reranker rescues / demotions	Decision rescues / demotions
DEV						
Baseline	ICD	46	11	97	34 / 57	83 / 16
Baseline	SNOMED	172	5	415	124 / 194	214 / 78
+ Pre-training	ICD	30	14	97	30 / 70	98 / 14
+ Pre-training	SNOMED	148	7	415	121 / 192	223 / 67
+ Post-training	ICD	45	8	75	77 / 24	31 / 14
+ Post-training	SNOMED	172	4	339	200 / 110	110 / 57
+ Pre + Post (final)	ICD	30	8	70	71 / 35	50 / 9
+ Pre + Post (final)	SNOMED	148	3	356	191 / 114	133 / 62
TEST						
Baseline	ICD	44	19	90	36 / 64	89 / 21
Baseline	SNOMED	240	9	439	136 / 227	249 / 86
+ Pre-training	ICD	28	14	81	33 / 76	115 / 14
+ Pre-training	SNOMED	215	8	440	136 / 222	266 / 79
+ Post-training	ICD	44	8	62	80 / 23	29 / 7
+ Post-training	SNOMED	240	5	350	215 / 127	124 / 47
+ Pre + Post (final)	ICD	28	11	64	69 / 40	62 / 13
+ Pre + Post (final)	SNOMED	215	6	371	212 / 131	143 / 52

Insights.

- With the original Qwen, the reranker demotes more than it rescues: on TEST, 227 SNOMED demotions against 136 rescues. With the trained reranker (final system) it flips to 212 rescues and 131 demotions.
- The decision layer is a net gain in every pipeline. Final system, TEST: SNOMED 143 rescues vs 52 demotions; ICD 62 vs 13.
- Pre-training cuts "gold not retrieved" errors: TEST ICD 44→28, SNOMED 240→215.

Table 20: SNOMED wrong-concept classes of the final outputs. Near miss = trusted qualifier/lexical neighbour of the gold.

Pipeline	Near miss	Same semantic tag	Same top level only	Different top level
DEV				
Baseline	10	240	50	292
+ Pre-training	13	235	45	277
+ Post-training	8	216	40	251
+ Pre + Post (final)	10	226	38	233
TEST				
Baseline	11	258	59	360
+ Pre-training	13	267	55	328
+ Post-training	10	243	50	292
+ Pre + Post (final)	10	254	43	285

Case 3 is a decision-layer rescue after a wrong raw ranking. Case 4 is a demotion after a correct raw ranking. Both failures had the gold in the pool, and both predictions are clinically close to the gold.

Table 21: Worked long-tail cases (final system, out-of-fold DEV; sampled with seed 13). Only public dictionary text is shown.

System / bucket	Gold	Prediction	Outcome	Raw reranker right?
ICD / rare	M19.011 Primary osteoarthritis, right shoulder	M19.011 Primary osteoarthritis, right shoulder	success	yes
SNOMED / rare	309538000 Numbness of foot	309538000 Numbness of foot	success	yes
ICD / rare	R53.81 Nervous debility	R53.81 Nervous debility	success	no
SNOMED / unseen	784580008 Estimated creatinine clearance	167181009 Measurement of renal clearance of creatinine	failure (wrong selection)	yes
SNOMED / rare	443694000 Uncontrolled type 2 diabetes mellitus	268519009 Diabetic - poor control	failure (wrong selection)	no

Coverage, confidence and abstention

TermMap always outputs one legal code per system (100% coverage, 0% abstention). Low confidence raises a *review flag*; it never removes the prediction. A confidence score is better at spotting a wrong pick than a missing gold [5], which matches the stress test below.

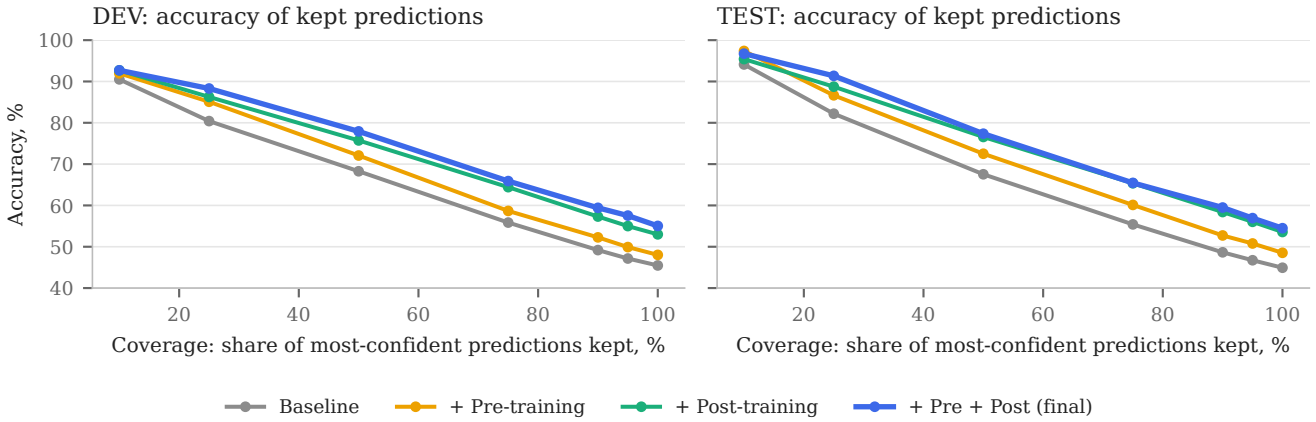


Figure 6: Risk–coverage (ICD and SNOMED pooled). If only the most confident 25% of predictions are kept, the final system is 91% accurate on TEST; at 50%, 77%.

Table 22: Review flag and calibration on TEST. Unflagged accuracy is the accuracy of predictions that are not flagged. Brier and ECE: lower is better. Missing-code AUROC: how well the missing-code score ranks pools without the gold.

Pipeline	System	Unflagged coverage	Unflagged accuracy	Flagged	Brier	ECE	Missing-code AUROC
Baseline	ICD	91.41	57.05 [51.81, 62.58]	8.59	0.1745	0.1104	80.89
Baseline	SNOMED	95.17	44.27 [41.35, 47.22]	4.83	0.1812	0.0373	76.40
+ Pre-training	ICD	93.87	66.34 [61.22, 71.48]	6.13	0.1688	0.1288	84.16
+ Pre-training	SNOMED	95.34	46.64 [43.53, 49.28]	4.66	0.1750	0.0540	78.63
+ Post-training	ICD	92.94	68.98 [63.45, 73.70]	7.06	0.1679	0.1310	83.91
+ Post-training	SNOMED	95.00	52.94 [50.00, 55.81]	5.00	0.1831	0.0549	79.00
+ Pre + Post (final)	ICD	93.87	71.90 [66.90, 76.82]	6.13	0.1645	0.1346	87.57
+ Pre + Post (final)	SNOMED	94.09	53.54 [50.57, 56.10]	5.91	0.1819	0.0496	80.04

Table 23: Missing-code stress test (final system, all DEV inputs): the gold is removed from every stage, so the right answer is impossible. Nearby = same ICD category or a trusted SNOMED neighbour.

System	n	Nearby prediction	Flag catches it	Nearby and flagged	Normal flag rate	False alarm (gold present)
ICD	327	46.48 [40.66, 51.89]	12.84 [9.48, 16.41]	2.75 [1.20, 4.59]	5.81 [3.35, 8.33]	3.37 [1.37, 5.75]
SNOMED	1041	2.88 [1.83, 4.04]	7.78 [6.02, 9.51]	0.19 [0.00, 0.49]	4.80 [3.48, 6.21]	2.58 [1.53, 3.85]

Reading. When the right code is missing, the engine often lands nearby for ICD (46%), but the review flag catches only

13% (ICD) and 8% (SNOMED) of these cases. Missing-code detection is a weak point (Section 6).

E Why Pre-training: Contrastive Adaptation of SapBERT

The problem it solves. Retrieval can only return codes whose names sit close to the mention in the embedding space. Chart text differs from dictionary text: "chronic kidney dis stg 3" vs "Chronic kidney disease, stage 3 unspecified". About half of the SNOMED evaluation rows have a code never seen in training, and new wording appears all the time. The embedding space itself has to place such mentions near the right concept. A richer space means a new term still gets a high similarity to its true concept.

How. Starting from SapBERT [1], we fine-tune only the *query* encoder with a contrastive (InfoNCE) loss: pull each training query towards its gold concept and push it away from *every* other concept in the dictionary (Figure 7). The dictionary vectors come from the original SapBERT and are *never updated*. Codes that never appear in training therefore keep exactly the vectors they had, and the adapted queries learn to find them. Queries are real chart mentions plus noisy dictionary names, so the encoder also learns typos, merged words and truncations.

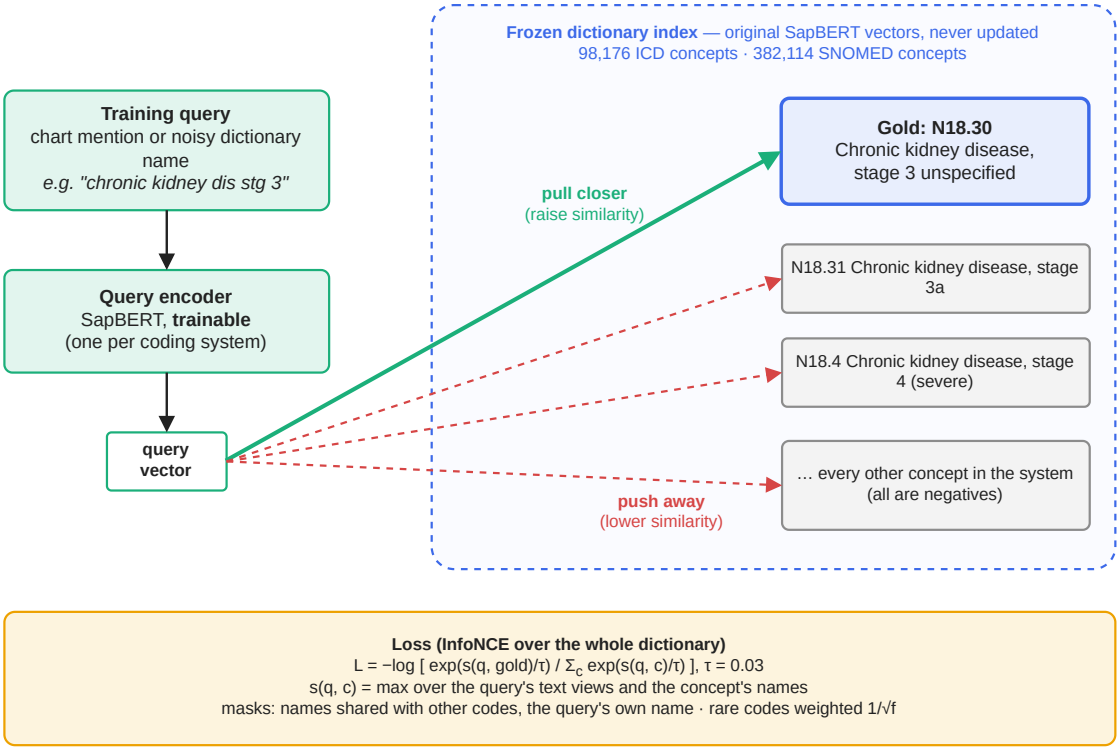


Figure 7: Contrastive pre-training with a frozen dictionary. Example strings are illustrative.

Table 24: Effect of pre-training on retrieval (DEV, final preprocessing). Original = off-the-shelf SapBERT queries. Adapted = our encoder. The adapted encoder had to pass all six rules per system to be kept: dense R@64 $\geq +1.0$; dense R@1 not lower; unseen R@64 ≥ -0.5; probe R@64 ≥ -1.0; full retrieval R@64 not lower; memory R@10 not lower. All passed.

	ICD			SNOMED		
Metric	Original	Adapted	Δ	Original	Adapted	Δ
Dense R@1	33.6	51.7	+18.0	37.0	44.7	+7.7
Dense R@10	62.4	78.6	+16.2	64.8	74.3	+9.4
Dense R@64	74.3	89.6	+15.3	78.2	87.1	+8.9
Dense R@64, unseen codes	77.8	84.0	+6.2	77.1	84.7	+7.6
Dense R@64, dictionary probe	94.7	96.9	+2.2	93.4	93.4	+0.0
Train-memory R@10	67.6	70.9	+3.4	41.5	44.5	+3.0
Full retrieval R@64	85.9	90.8	+4.9	83.5	85.8	+2.3

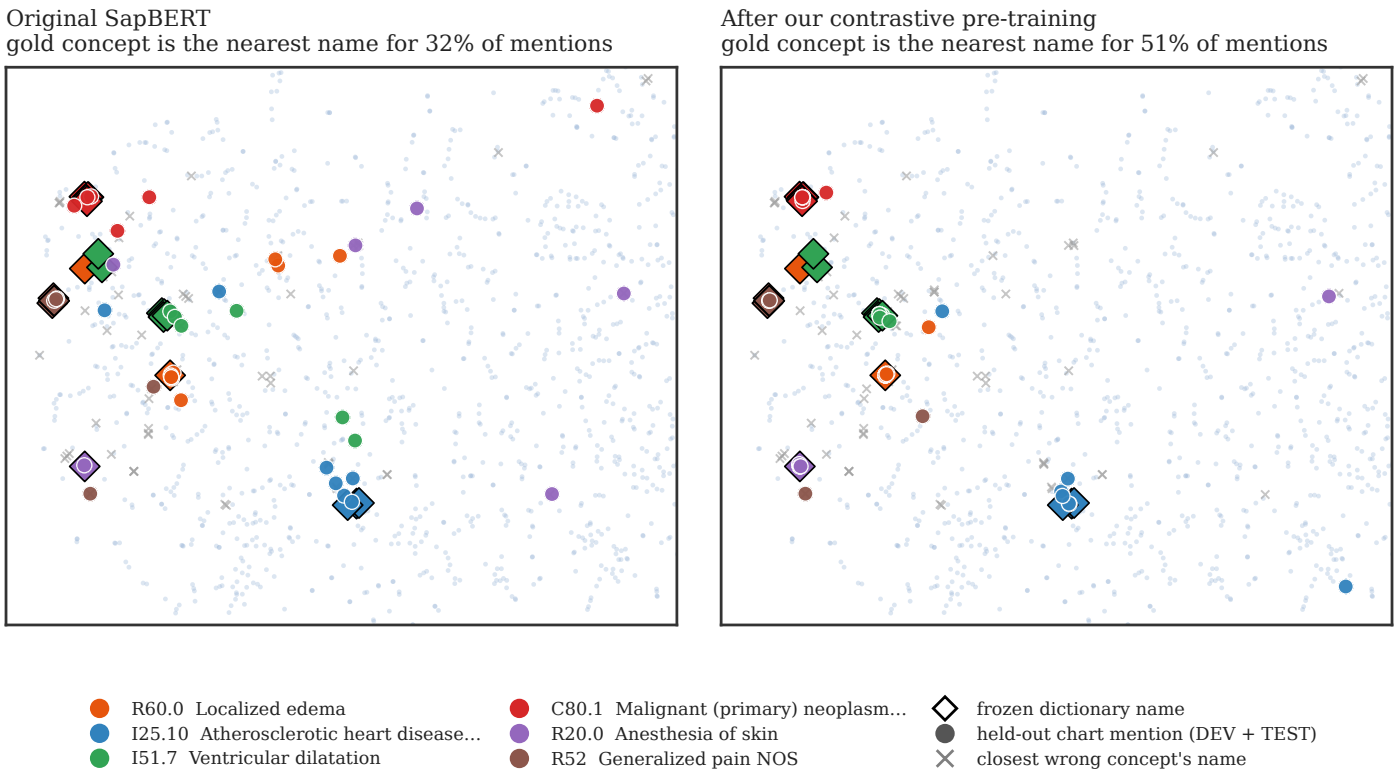


Figure 8: **What pre-training does to the embedding space** (our data, in the style of the SapBERT paper [1]). Both panels show the same map of frozen dictionary names (t-SNE, light dots). Diamonds are the names of six ICD concepts: the most-mentioned held-out concepts, skipping catch-all "Other . . . " codes. Circles are DEV and TEST chart mentions of those concepts, placed among their five most similar dictionary names. With the original encoder (left) the mentions are scattered, often next to wrong concepts (grey crosses). After contrastive pre-training (right) most of them sit on their own concept. Over all 651 held-out ICD mentions, the gold concept is the single nearest name for 32% before and 51% after.

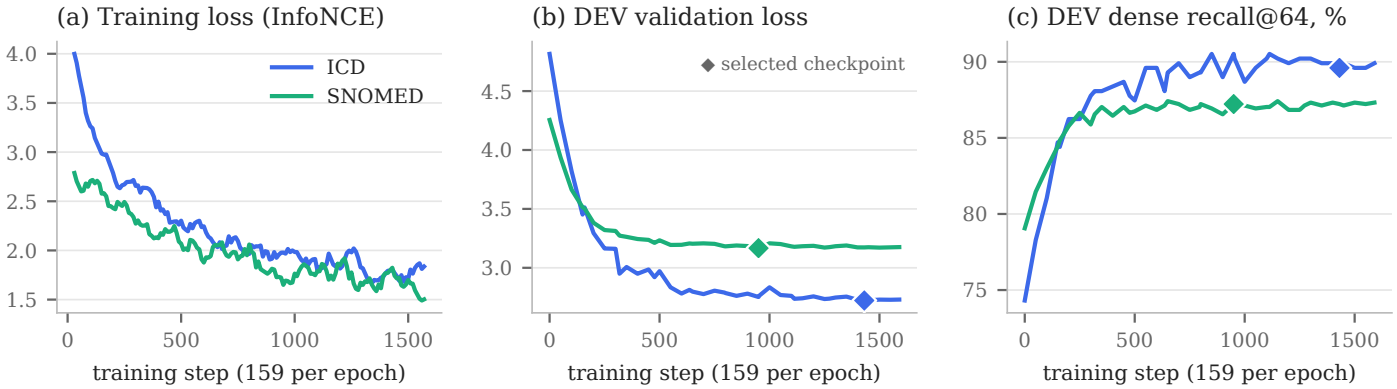


Figure 9: Training curves. (a) InfoNCE training loss (smoothed). (b) DEV validation loss; diamonds mark the selected checkpoint per system (lowest DEV loss). (c) DEV dense recall@64. Recall stops improving after about 4–5 epochs; picking the lowest-loss checkpoint did not cherry-pick a recall peak.

Table 25: Why v1 was abandoned and what v2 changed.

	v1 (abandoned)	**v2 (final)**
Method	SapBERT-style multi-similarity loss on synonym pairs; one encoder for dictionary and queries	InfoNCE over all concepts; query encoder only; dictionary frozen
What moved	Every dictionary vector (cosine to the original 0.43 ICD, 0.57 SNOMED), including codes never seen in training	Only query and training-memory vectors
Result	Full-pipeline DEV recall@64 84.1→81.7%; failed its acceptance gate	Dense recall@64 ICD +15.3, SNOMED +8.9; all six rules pass

F Why Post-training: the Reranker and the Decision Layer

Why we need a reranker at all. Retrieval often finds the gold but does not put it first. On TEST, the final retriever has the ICD gold in its top 64 for 91.4% of inputs, but at rank 1 for only 44.5%. The retriever sees the mention alone. Choosing between close codes needs the note: "MS" in a neurology history is multiple sclerosis, not mitral stenosis; "no chest pain" carries a negation; "left knee" sets laterality. The reranker is the only model that reads mention, context and a full description of each candidate together, as in context-aware linking [4] and LLM ranking over alias retrieval [3].

Why the reranker needs post-training. Qwen3-Reranker is a general-purpose relevance model. It has never seen ICD/SNOMED coding conventions: which code coders choose for a mention, how specific to be, or how headers relate to billable children. Without post-training it lowered top-1 accuracy compared with retrieval in every case (TEST: ICD 40.8→32.2%, SNOMED 36.7→29.1%). With the original reranker, the full pipeline maps only 15% of held-out ICD dictionary names correctly. LoRA post-training on our own retriever's hard negatives teaches these conventions: raw ICD top-1 rises to 58.3% and held-out-name accuracy to 61%. Only 1.7% of the weights are trained, which fits on one GPU and keeps the base model's language knowledge.

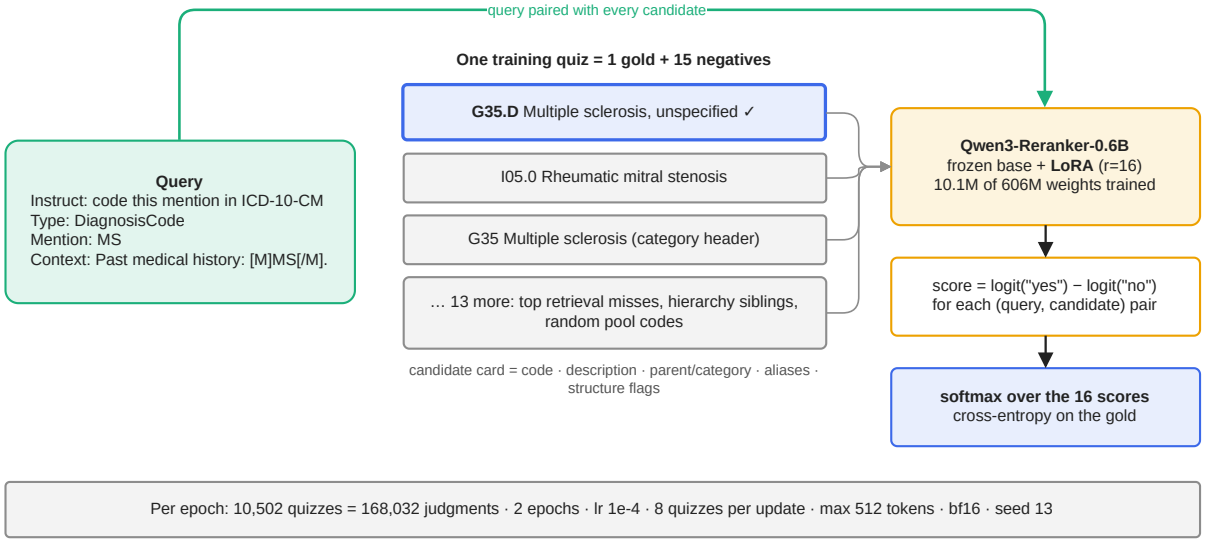


Figure 10: One post-training quiz. Negatives come from our own retrieval pool (hardest misses, hierarchy siblings, random pool codes), so the model learns exactly the confusions it will face. Candidate order and the visibility of the type field are seeded.

Why a decision layer after the reranker. Even the trained reranker makes mistakes that other signals can catch: exact dictionary matches, training memory, the type prior and the specificity rules. The decision layer is a small logistic model over all these features, fitted on DEV. It gives the largest stage gain in every pipeline (Tables 8–9) and has more rescues than demotions (Table 19).

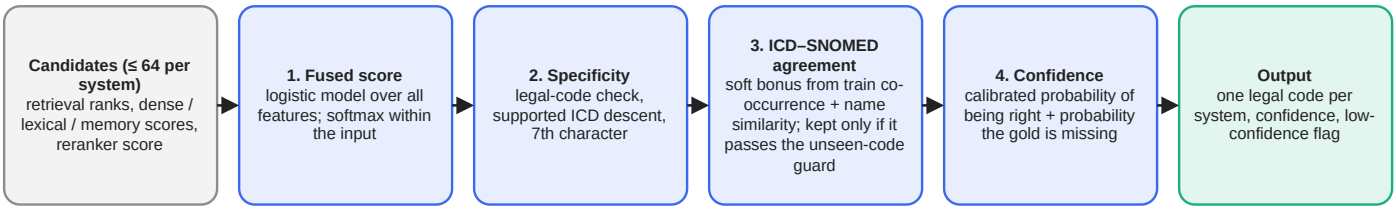


Figure 11: Decision layer, applied per system.

Checkpoint choice. The automatic selector picked LoRA epoch 1 on its original DEV diagnosis-only criterion (53.9%). We explicitly chose epoch 2 for the final comparison. Both adapters are kept, and the choice is recorded in the release metadata.

G Pipeline Details: Preprocessing, Hierarchy and Retrieval

Preprocessing

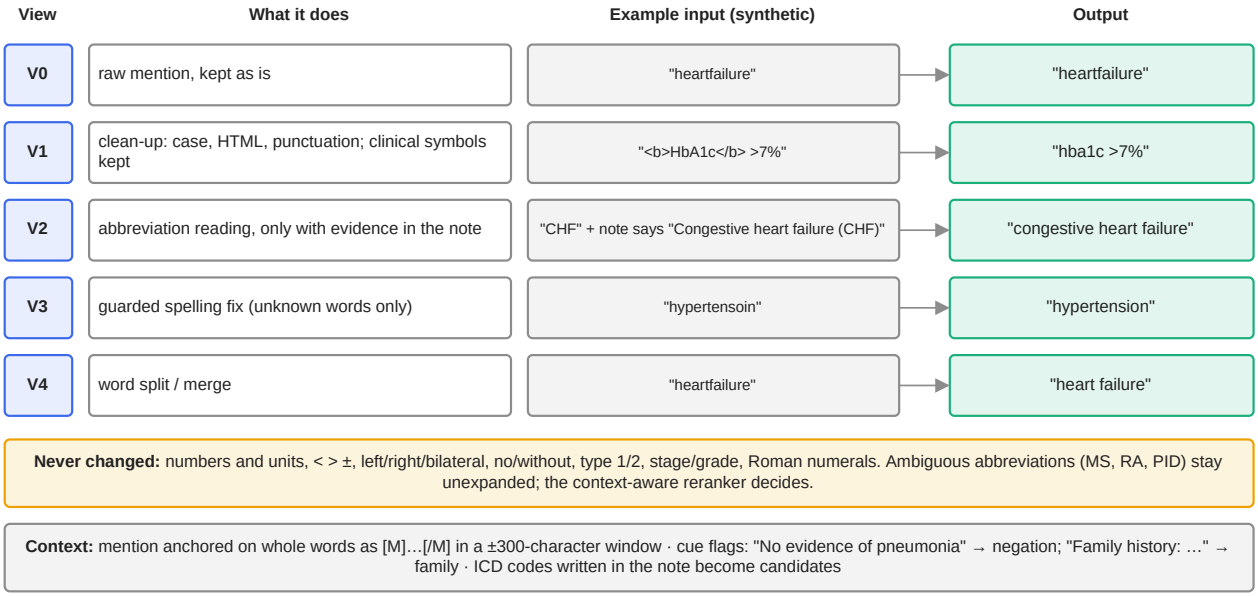


Figure 12: The five mention views. Every output shown was produced by the final preprocessing code on the synthetic input. Each later view is only added when its rule has evidence; otherwise it equals the cleaned view.

Terminology and hierarchy

Table 26: Terminology layer.

ICD-10-CM codes	98,182
billable leaves	74,718
ICD names searched (dictionary + abbreviations)	159,132
CDC index/inclusion names (built, disabled in final retrieval)	68,369
SNOMED CT US concepts	382,124
with an official semantic tag	382,120
SNOMED names searched	999,771

ICD. A code's parent is the longest existing prefix (ignoring the dot). Chapters and blocks come from the CDC FY2026 Tabular XML, which also gives 7th-character meanings. Nodes are stored in pre-order with an interval (in, out) per code, kept in SQLite: "is a under b" is $O(1)$, and a subtree is one contiguous slice.

SNOMED. The FSN semantic tag (matched against the official tag list) maps to its top-level hierarchy. Qualifier siblings (left/right, acute/chronic, "history of", "no") are found by lexical rules. Concepts sharing a core name form contiguous "families". These links are approximate, not the official IS-A graph.

Search cost. Dense search is one exact matrix product over all name vectors on the GPU, in chunks. Lexical search uses sparse character n-gram and BM25 indexes. Fusion and expansion take 23 ms per input (median).

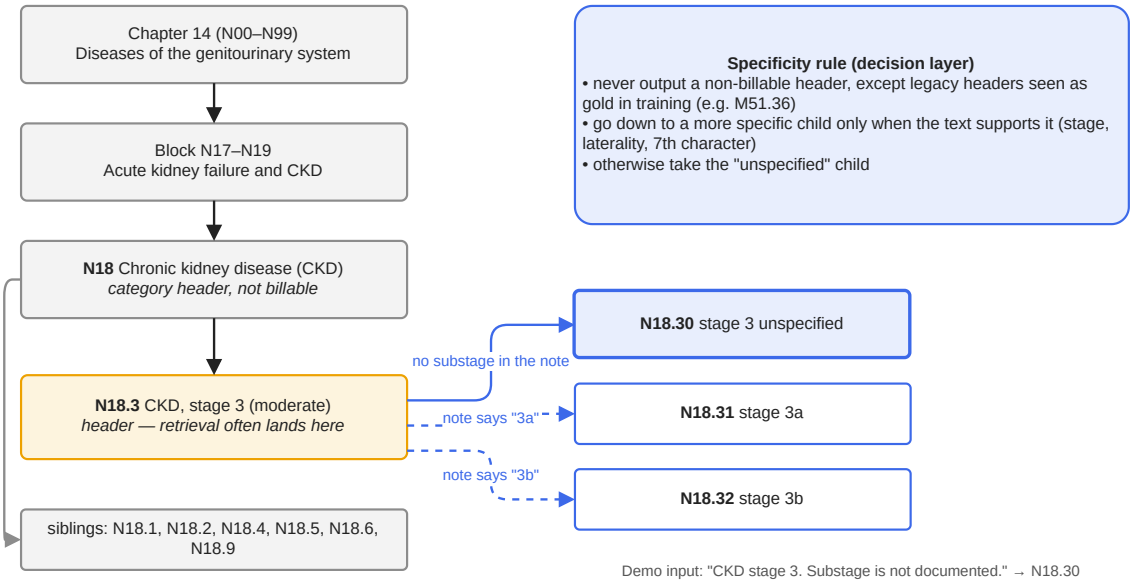


Figure 13: How the ICD hierarchy drives specificity. Retrieval often matches a header. The decision layer moves down only when the text supports a more specific child.

Table 27: Retrieval settings of the final system.

Channels	exact / normalised exact, character 3–4-gram TF-IDF, BM25, dense SapBERT, training-mention memory, ICD codes written in the note
Per-channel depth / fusion	100 candidates each; reciprocal-rank fusion k=60; keep 50
Expansion	ICD subtree children and look-alike siblings; SNOMED qualifier siblings; pool capped at 64
Query views used	ICD: V0–V2; SNOMED: V0–V4
Disabled in the retained policy	CDC index names as search aliases, SNOMED→ICD bridge, SNOMED family expansion
Encoders	dictionary names: original SapBERT; queries and training memory: adapted encoder of that system; type classifier: original SapBERT

Time complexity

N = concepts, A = dictionary names, d = embedding width, s = nodes in the subtree read, h = tree height, B = cost of one SapBERT encoding. Only the highest-order term is shown.

Table 28: Time complexity of TermMap's key operations and of dictionary updates, compared with a simple scan or rebuild.

Operation	**TermMap**	**Simple approach**
Build ICD prefix tree and subtree intervals	$O(N \log N)$	$O(N^2)$
Code / parent / depth / tag lookup	$O(1)$	$O(N)$
ICD ancestor check ("is a under b")	$O(1)$	$O(h)$
Read an ICD subtree	$O(s)$	$O(N)$
Chunked exact dense scoring (per query vector)	$O(Ad)$	$O(Ad)$
Dictionary updates (no model retraining needed)		
Add or edit names, same codes	$O(AB)$	$O(AB)$
Add or edit ICD codes	$O(N \log N)$	$O(N^2)$
Add or edit SNOMED concepts or tags	$O(A \log A)$	$O(N^2)$

The SNOMED update cost assumes the lexical links built between matching concept pairs are far fewer than the names.

H Alternative Solution: a Tool-Using LLM Agent

Idea. Alongside the main pipeline we built and evaluated a separate approach in which a large language model works like a coder. It searches the dictionaries, reads code descriptions, looks at ICD parents and children and at similar training mentions, and then commits to one code per system. The problem statement suggests using an LLM "where judgement is needed". This workstream tests how far that goes on its own, without a trained reranker; agentic reranking has also been evaluated for SNOMED linking [4]. It was kept separate from the main engine and is not part of the submitted pipeline.

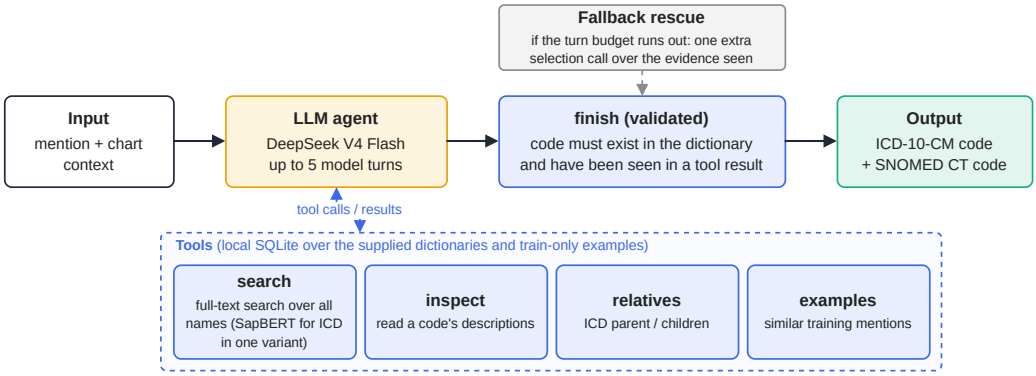


Figure 14: Agent design. All tools run locally over the supplied dictionaries and train-only examples. A *finish* is accepted only if the code is in the dictionary and appeared in a tool result. If the turn budget runs out, one extra selection call chooses from the evidence already seen (fallback rescue).

Table 29: Main agent results on 500 diverse diagnosis inputs (each has an ICD and a SNOMED gold). Exact-code accuracy, %. A–C are fixed-evidence selectors over the agent's saved tool results.

Configuration	ICD	SNOMED	Both	Reading
Original text search/inspection	67.4	70.4	53.2	Full paired reference
SapBERT ICD + original SNOMED	69.2	69.2	53.0	Mixed result; not a proven overall upgrade
A: compact fixed-evidence selector	65.4	69.6	48.8	Not an agent replacement
B: A plus coding/task guide	67.4	70.2	50.8	Some improvement over A
C: B plus lexical candidate distinctions	68.8	69.6	51.2	ICD gain with SNOMED losses
Fallback-only B rescue after failed investigation	**70.6**	**71.4**	**53.6**	Controlled continuation; 465 accepted decisions preserved

Table 30: Agent long-tail accuracy (%) on the same 500 inputs. Unseen = code absent from the agent's training-example store. Highlighted cells are ≥60%.

System / training frequency	Inputs	Original agent	SapBERT ICD variant	Variant + rescue
ICD unseen	114	56.1	58.8	59.6
ICD rare (15 examples)	261	**69.7**	**72.0**	**73.2**
ICD common (6+)	125	**72.8**	**72.8**	**75.2**
SNOMED unseen	241	58.5	57.3	**60.2**
SNOMED rare	229	**80.3**	**79.5**	**80.8**
SNOMED common	30	**90.0**	**86.7**	**90.0**

Table 31: Other agent experiments (ICD/SNOMED accuracy unless stated). Sample sizes and protocols differ, so compare only within a row.

Experiment	Result and decision
Initial smoke and model/tool checks	Agent scored 70%/70% on 20 inputs. On a separate 40 inputs, DeepSeek search/inspection scored 82.5%/80%, ahead of the smaller Qwen comparator. Small samples did not predict the later 500-input result.
Query recovery, mandatory comparison, extra prompt checks	On 80 inputs, none established an overall improvement over the 65%/71.25% baseline; some added failures or regressed correct choices.
Dense retrieval and selective deeper reading	ICD-only change reached 70%/70% on 80 inputs. Selective SNOMED review did not help. The subsequent 500-input comparison remained inconclusive.
Retrieval cascade and preprocessing	On a separate 500-input retrieval benchmark, SapBERT achieved 64.6% ICD recall@12; the exact-first cascade achieved 80.6% SNOMED recall@12. Combined preprocessing did not improve those figures. These are candidate-recall scores, not mapping accuracy.
Higher reasoning	On 100 inputs, off scored 60%/73%; high scored 58%/63%, with 13 truncated outputs and 2.63× the cost. No expansion.
Lexical contextual/contrast memory	On 100 inputs, mention/context/contrast scored 61%/71%, 62%/72%, and 61%/74%. Repeated-call variation on identical requests undermined attribution of small gains.
Semantic contextual memory	On the same 100-input pilot, semantic memory scored 60%/70% against a cached lexical control's 62%/72%. It retrieved different cases but did not improve selection. No expansion.

Table 32: Selector pilots on 100 inputs (number correct out of 100). Compare within each block only.

Condition	ICD	SNOMED	Both
Reasoning off, 8,192-token common cap	60	73	45
High reasoning, same evidence/cap	58	63	45
Mention-based lexical memory	61	71	48
Contextual lexical memory	62	72	49
Lexical contrast memory	61	74	50
Semantic experiment: canonical cached lexical control	62	72	49
Semantic case retrieval	60	70	47

What we learned.

- **The agent is strong on diagnoses.** Its best run reaches 70.6% ICD and 71.4% SNOMED on diagnosis inputs, and about 60% on unseen codes in both systems. For SNOMED this is higher than the main pipeline's diagnosis accuracy (62.05% on TEST). The samples differ, though: the agent's 500 inputs are reused development cases, diagnoses only. The figures show the agent is competitive; they do not show it is better.
- **Retrieval and selection both limit it.** After rescue, 70 ICD and 65 SNOMED errors never saw the gold code, and 77 / 78 saw it but chose wrongly. Perfect selection from the saved evidence would reach only 86% / 87%.
- **More retrieval did not mean more accuracy.** SapBERT exposed the ICD gold in 99 more first searches but gave only 9 more correct answers, because later agent searches already found many of them.
- **Generic prompt tricks did not help.** Higher reasoning, extra instructions, query recovery and contextual memory gave no reliable gain. Reliable finalisation (the fallback rescue) was the clearest targeted improvement.
- **Cost and determinism.** Each input needs several hosted LLM calls (about $1.64 of new API spend for the 500-input comparison), and hosted outputs are not bit-identical across reruns. The main engine is local, deterministic and needs one reranker pass per system. This is why the trained reranker is the submitted design. The agent is a candidate for low-confidence inputs only, behind the review flag (Section 6).

Models and data handling: DeepSeek V4 Flash (and Qwen3.5-9B in an early comparison) through hosted inference APIs (OpenRouter with DeepInfra/CoreWeave). The dictionaries and training examples stayed in a local SQLite database, but each input's mention, chart context and tool results were sent to the hosted model. This is another reason the submitted engine runs fully locally.

I Latency and Deliverables

Where the time goes

Fine-grained timing of the final system on one RTX A6000 (6 CPU cores, 48 GiB RAM): 50 seeded DEV inputs, run one at a time after warm-up. Each input predicts *both* systems. Query vectors, lexical and dense scores and reranker scores are recomputed for every input (no query or score caches). Output validation is included.

Table 33: Warm latency per input by component (ms). Share = component's share of the mean request time; component medians do not add up to the total median.

Component	Median	p95	Share of time (%)
Mention preprocessing, context and type handling	10.0	12.2	0.7
SapBERT query encoding: all three encoder roles	59.2	61.4	3.6
Character TF-IDF and BM25 scoring	949.2	1009.2	56.5
Dense alias search: both systems / configured variants	25.2	29.3	1.6
Mention-memory scoring and query bookkeeping (residual)	10.8	11.6	0.7
Candidate fusion, ranking and configured structural expansion	28.0	478.0	4.8
Contextual Qwen LoRA reranking: both systems	461.1	579.3	30.8
Decision features, fusion, specificity, coupling and confidence	18.1	30.1	1.3
Output assembly and code/schema validation	1.5	1.7	0.1
Complete warm request	**1567.1**	**1965.1**	100

Table 34: Nested sub-components (already inside the rows above; ms).

Sub-component	Median	p95
Original SapBERT: type-inference query vectors	20.60	22.10
Adapted ICD SapBERT: ICD query vectors	19.18	21.16
Adapted SNOMED SapBERT: SNOMED query vectors	19.03	19.59
Character TF-IDF scoring	845.84	896.24
BM25 scoring	101.46	115.87
ICD alias dense search, including prepared CDC variants	6.66	9.80
SNOMED alias dense search	18.30	21.70

Reading. Character n-gram and BM25 scoring take 56.5% of request time and the reranker 30.8%; all three encoders together take only 3.6%. The fastest win is therefore the lexical channel, not the neural models. An earlier, coarser benchmark gave 1.35 s median / 1.76 s p95 because it stopped the timer before output validation. Peak GPU memory was 39.5 GiB (including the vLLM reservation) and the mean service time of 1.53 s gives about 2,356 inputs per hour on one GPU. Sustained multi-request throughput has not been load-tested.

Deliverables

Table 35: Required deliverables.

Deliverable	Where
Working end-to-end mapping engine	`run.sh` and `scripts.inference.predict` in the repository
Predictions in the required schema	Produced by `run.sh`; validated (legal codes, order, string IDs)
Private GitHub repository with judge access	`https://github.com/kingslayer35/HackLabs_IITR`
README, `.env.example`, single command	Repository root
Trained models and private artifacts	Private model repositories and a restricted artifact bundle (links in the README)
Documentation PDF	This report
Demo video (2–3 minutes)	Link in the README